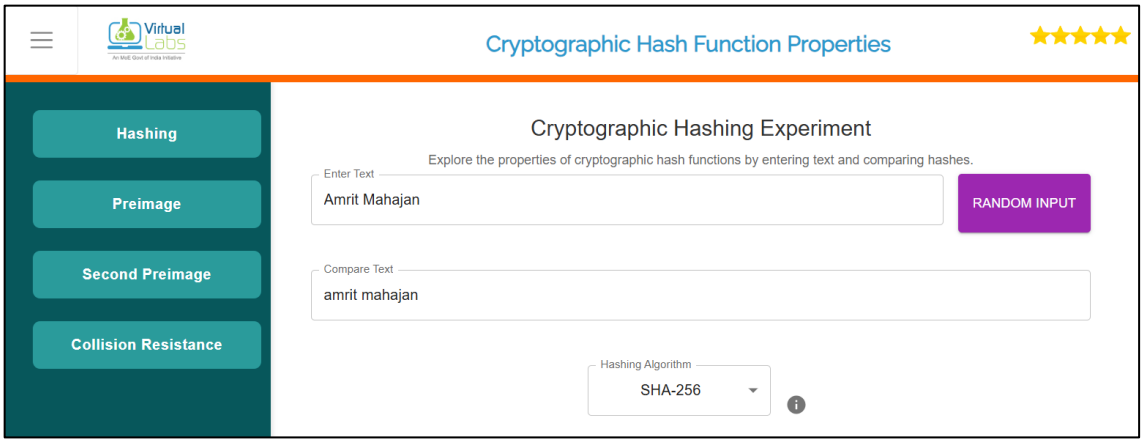
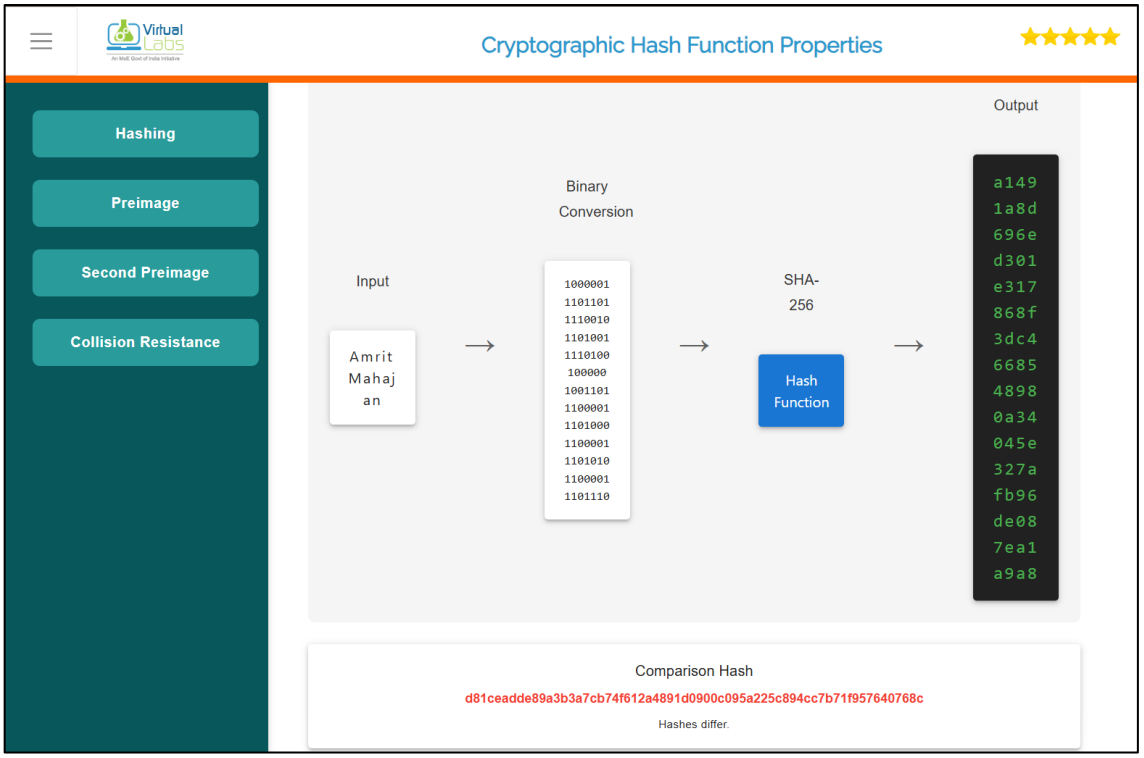
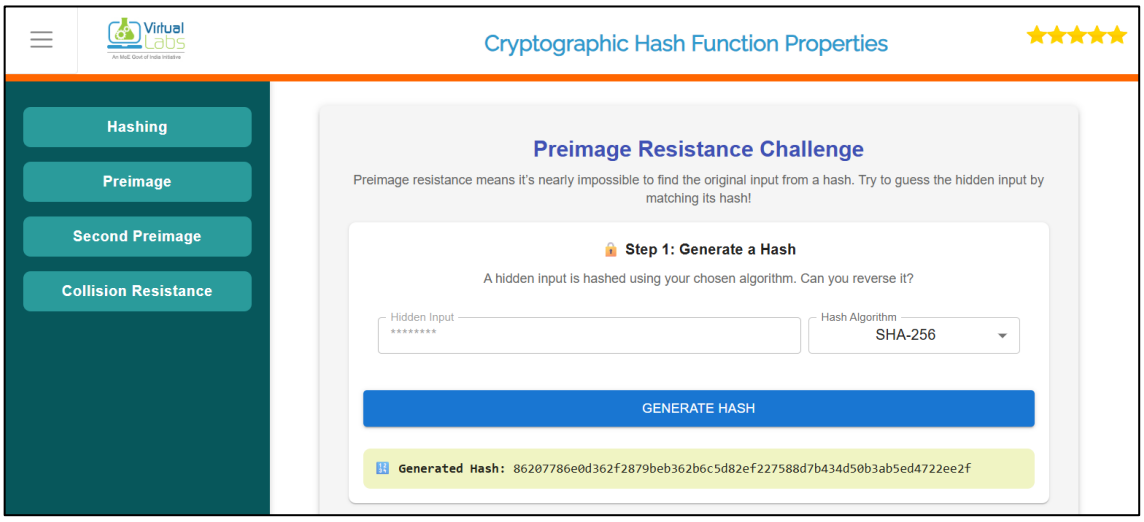
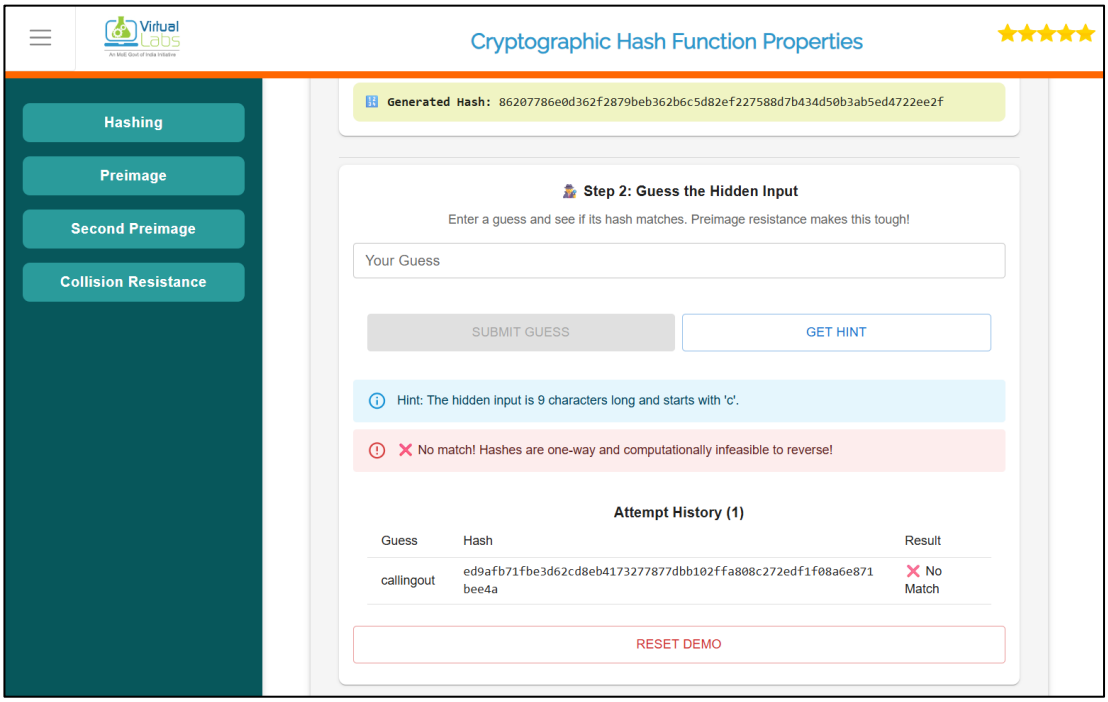
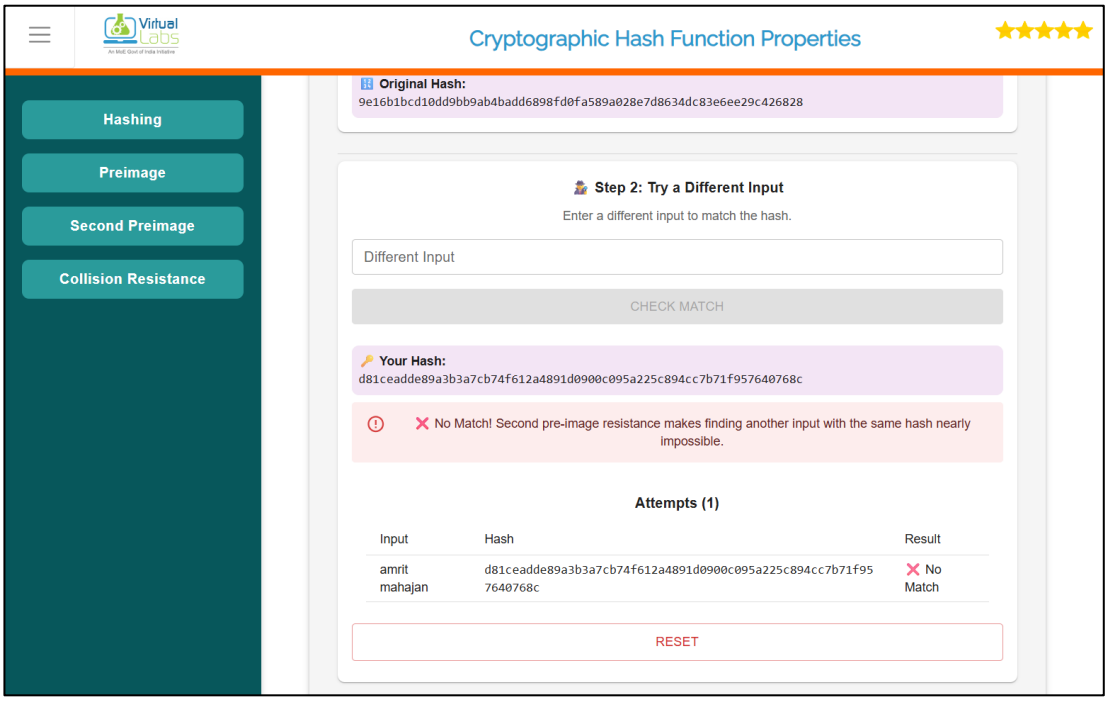
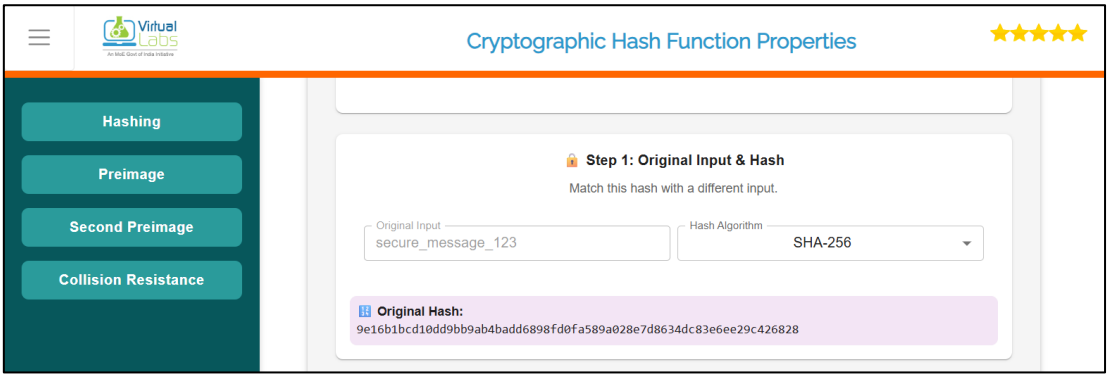

Virtual Labs
Cryptographic Hash Function Properties
Hashing
Preimage
Second Preimage
Collision Resistance
Cryptographic Hashing Experiment
Explore the properties of cryptographic hash functions by entering text and comparing hashes.
Enter Text
Amrit Mahajan
RANDOM INPUT
Compare Text
amrit mahajan
Hashing Algorithm
SHA-256


Virtual Labs
Cryptographic Hash Function Properties
Hashing
Preimage
Second Preimage
Collision Resistance
Input
Amrit Mahajan
Binary Conversion
1000001 1101101 1110010 1101001 1110100 100000 1001101 1100001 1101000 1100001 1101010 1100001 1101110
SHA-256
Hash Function
Output
a149 1a8d 696e d301 e317 868f 3dc4 6685 4898 0a34 045e 327a fb96 de08 7ea1 a9a8
Comparison Hash
d81ceadde89a3b3a7cb74f612a4891d0900c095a225c894cc7b71f957640768c
Hashes differ.


Virtual Labs
Cryptographic Hash Function Properties
Hashing
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Preimage Resistance Challenge
Preimage resistance means it's nearly impossible to find the original input from a hash. Try to guess the hidden input by matching its hash!
Step 1: Generate a Hash
A hidden input is hashed using your chosen algorithm. Can you reverse it?
Hidden Input
Hash Algorithm
SHA-256
GENERATE HASH
Generated Hash: 86207786e0d362f2879beb362b6c5d82ef227588d7b434d50b3ab5ed4722ee2f


Virtual Labs
Cryptographic Hash Function Properties
Hashing
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Second Preimage
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Generated Hash: 86207786e0d362f2879beb362b6c5d82ef227588d7b434d50b3ab5ed4722ee2f
Step 2: Guess the Hidden Input
Enter a guess and see if its hash matches. Preimage resistance makes this tough!
Your Guess
SUBMIT GUESS
GET HINT
Hint: The hidden input is 9 characters long and starts with 'c'.
No match! Hashes are one-way and computationally infeasible to reverse!
Attempt History (1)
Guess
Hash
Result
callingout
ed9afb71fbe3d62cd8eb4173277877dbb102ffa808c272edf1f08a6e871bee4a
No Match
RESET DEMO


Virtual Labs
Cryptographic Hash Function Properties
Hashing
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Second Preimage
Collision Resistance
Original Hash:
9e16b1bcd10dd9bb9ab4badd6898fd0fa589a028e7d8634dc83e6ee29c426828
Step 2: Try a Different Input
Enter a different input to match the hash.
Different Input
CHECK MATCH
Your Hash:
d81ceadde89a3b3a7cb74f612a4891d0900c095a225c894cc7b71f957640768c
No Match! Second pre-image resistance makes finding another input with the same hash nearly impossible.
Attempts (1)
Input
Hash
Result
amrit mahajan
d81ceadde89a3b3a7cb74f612a4891d0900c095a225c894cc7b71f957640768c
No Match
RESET


Virtual Labs
Cryptographic Hash Function Properties
Hashing
Preimage
Second Preimage
Collision Resistance
Step 1: Original Input & Hash
Match this hash with a different input.
Original Input
secure_message_123
Hash Algorithm
SHA-256
Original Hash:
9e16b1bcd10dd9bb9ab4badd6898fd0fa589a028e7d8634dc83e6ee29c426828